

AI-Powered Telecom Customer Churn Prediction and Retention Strategy

Abstract

Customer churn is one of the biggest challenges faced by telecom companies. Acquiring a new customer is considerably more expensive than retaining an existing one, making early churn prediction a critical business objective. This project develops a machine learning-based customer churn prediction system using CatBoost to identify customers likely to leave the service. The model classifies customers into Low, Medium, and High Risk categories based on their predicted churn probability, enabling targeted retention strategies. The final CatBoost model achieved a ROC-AUC score of approximately **0.702**, demonstrating reliable predictive performance while generating actionable business insights.

1. Introduction

The telecommunications industry operates in a highly competitive environment where customer retention directly impacts profitability. Customers may discontinue services for several reasons including poor service quality, changing usage patterns, pricing dissatisfaction, or better offers from competitors.

Traditional approaches identify churn only after customers leave, resulting in lost revenue. Machine Learning enables proactive identification of customers at risk of churning, allowing businesses to intervene before customer loss occurs.

The objective of this project is to build an intelligent churn prediction system capable of identifying high-risk customers and supporting data-driven retention strategies.

2. Business Problem

The primary business challenge is reducing customer churn while maximizing customer lifetime value.

The company seeks to:

- Predict customers likely to churn.
 - Understand factors influencing churn.
 - Segment customers according to churn risk.
 - Recommend business actions for customer retention.
 - Improve revenue by reducing customer attrition.
-

3. Objectives

The project objectives include:

- Perform comprehensive Exploratory Data Analysis.
- Clean and preprocess telecom customer data.

- Engineer meaningful business features.
 - Train an advanced CatBoost classifier.
 - Evaluate model performance using ROC-AUC.
 - Identify the most influential features.
 - Categorize customers into Low, Medium, and High Risk groups.
 - Generate business recommendations based on model predictions.
-

4. Dataset Description

The telecom dataset contains approximately **100,000 customer records** with behavioral, demographic, financial, and service usage information.

Major feature categories include:

- Customer demographics
- Monthly revenue
- Voice usage
- Data usage
- Contract duration
- Device information
- Service usage statistics
- Customer activity metrics

Target Variable:

- **Churn**
 - 1 = Customer left
 - 0 = Customer retained
-

5. Data Preprocessing

The following preprocessing steps were performed:

- Handling missing values
- Removing duplicate records
- Encoding categorical variables
- Feature scaling where necessary
- Data validation

- Train-test split using stratified sampling
-

6. Exploratory Data Analysis

EDA was conducted to understand customer behavior and identify relationships with churn.

The analysis included:

- Missing value analysis
- Distribution plots
- Correlation analysis
- Feature importance exploration
- Customer behavior trends
- Revenue analysis
- Usage analysis

Key findings indicated that customer usage patterns and behavioral changes are stronger indicators of churn than demographic information alone.

7. Feature Engineering

Several business-oriented features were created to improve predictive performance.

Examples include:

- Revenue Stability
- Revenue Change Ratio
- Device Cost per Day
- Device Age Efficiency
- Customer Usage Ratios
- Behavioral Change Metrics

Feature engineering increased the model's ability to identify subtle churn patterns.

8. Machine Learning Model

Among multiple algorithms evaluated, **CatBoost Classifier** was selected due to its ability to handle complex tabular datasets and categorical relationships.

Model Parameters:

- Algorithm: CatBoost Classifier
- Evaluation Metric: ROC-AUC

- Learning Rate: 0.03
 - Depth: 8
 - Iterations: 2000
 - Random Seed: 42
-

9. Model Evaluation

Performance Metrics:

- ROC-AUC: **0.702**
- Accuracy: ~64%
- Precision: ~64%
- Recall: ~64%
- F1 Score: ~64%

ROC-AUC was selected as the primary evaluation metric because it provides robust performance evaluation for binary classification problems.

10. Feature Importance

The CatBoost model identified several highly influential features, including:

- Months with the company
- Change in usage
- Average monthly usage
- Equipment payment duration
- Total recurring charges
- Revenue change
- Device price
- Device age ratio

These variables strongly influence customer churn behavior.

11. Customer Risk Segmentation

Customers were divided into three business-oriented risk groups.

Risk Level Customers Churn Rate

Low	18,595	8.16%
-----	--------	-------

Risk Level Customers Churn Rate

Medium 65,605 50.94%

High 15,800 92.56%

The High Risk segment exhibits an extremely high probability of churn and should be prioritized for retention efforts.

12. Business Insights

Key findings include:

- High-risk customers have a **92.56%** churn rate.
 - Customer usage decreases as churn risk increases.
 - High-risk customers continue to generate significant revenue, making retention financially valuable.
 - Usage behavior is a stronger predictor of churn than customer tenure.
 - Medium-risk customers represent the largest customer segment and offer the greatest opportunity for reducing overall churn.
-

13. Business Recommendations

Recommended actions include:

High Risk

- Immediate customer outreach
- Personalized retention offers
- Discount campaigns
- Loyalty incentives
- Dedicated customer support

Medium Risk

- Personalized marketing campaigns
- Usage reminders
- Promotional bundles
- Customer engagement programs

Low Risk

- Continue standard customer service
- Reward customer loyalty

- Monitor behavioral changes
-

14. Business Impact

By identifying customers before they churn, the company can significantly reduce customer attrition.

For example, retaining only **30% of High Risk customers** could preserve approximately **4,740 customers**, improving recurring revenue while reducing acquisition costs.

15. Future Work

Future improvements include:

- Hyperparameter optimization
 - XGBoost and LightGBM ensemble models
 - Explainable AI using SHAP values
 - Real-time churn prediction dashboards
 - Automated customer retention recommendation systems
 - Deployment using Flask/FastAPI
 - Cloud deployment with Docker
-

16. Conclusion

This project successfully developed an end-to-end telecom churn prediction system using CatBoost. The model achieved a ROC-AUC score of approximately **0.702**, effectively identifying customers likely to churn. Risk segmentation enabled clear business prioritization, while feature engineering improved predictive performance. The resulting solution provides telecom companies with actionable insights to proactively retain customers, reduce churn, and improve long-term profitability.